LLM Hallucination: Theory, Experiments, and Limits

Based on the 2026 Comprehensive Survey + Empirical Analysis

MACHINE LEARNING RESEARCH | GRADUATE SEMINAR | 2026

Why Hallucination Matters

LLMs generate fluent, confident text even when factually incorrect — a dangerous property in high-stakes domains.

Healthcare

Fabricated drug interactions or diagnoses could cause direct patient harm.

Finance & Law

Invented case citations or market data may drive consequential decisions.

Core Problem

Fluency creates an illusion of reliability, masking factual errors from users.

⚠ **Key Insight:** "LLMs sound confident even when wrong." Calibration failure is the fundamental risk.

Defining Hallucination

Formal Definition

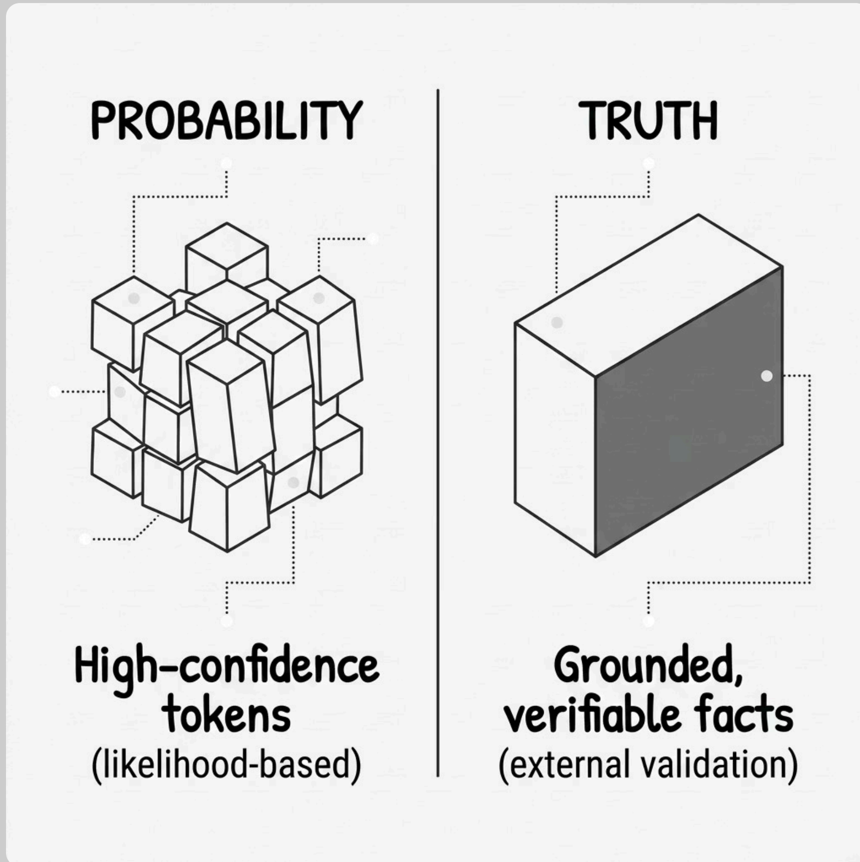
A **hallucination** is generated content that is (a) nonsensical or (b) unfaithful to the provided source context — yet presented with fluent, confident language.

Probabilistic Explanation

LLMs model $P(x_t|x_{1:t-1})$ — the next-token distribution. The objective is **likelihood maximization**, not truth verification. High-probability tokens need not correspond to factual content.

Key Insight: LLMs predict tokens, not truth. Fluency and factuality are orthogonal objectives.

Probability vs. Truth



Types of Hallucination

The 2026 survey establishes a multi-dimensional taxonomy. Two orthogonal axes are most analytically useful.

Dimension	Category A	Category B
Source	**Intrinsic** — Contradicts input context (e.g., misquotes provided text)	**Extrinsic** — Adds unsupported external claims (e.g., invents a citation)
Content Type	**Factual** — Verifiable real-world errors (e.g., wrong date, person)	**Faithfulness** — Deviates from source material in summarization tasks

Example — Intrinsic Factual

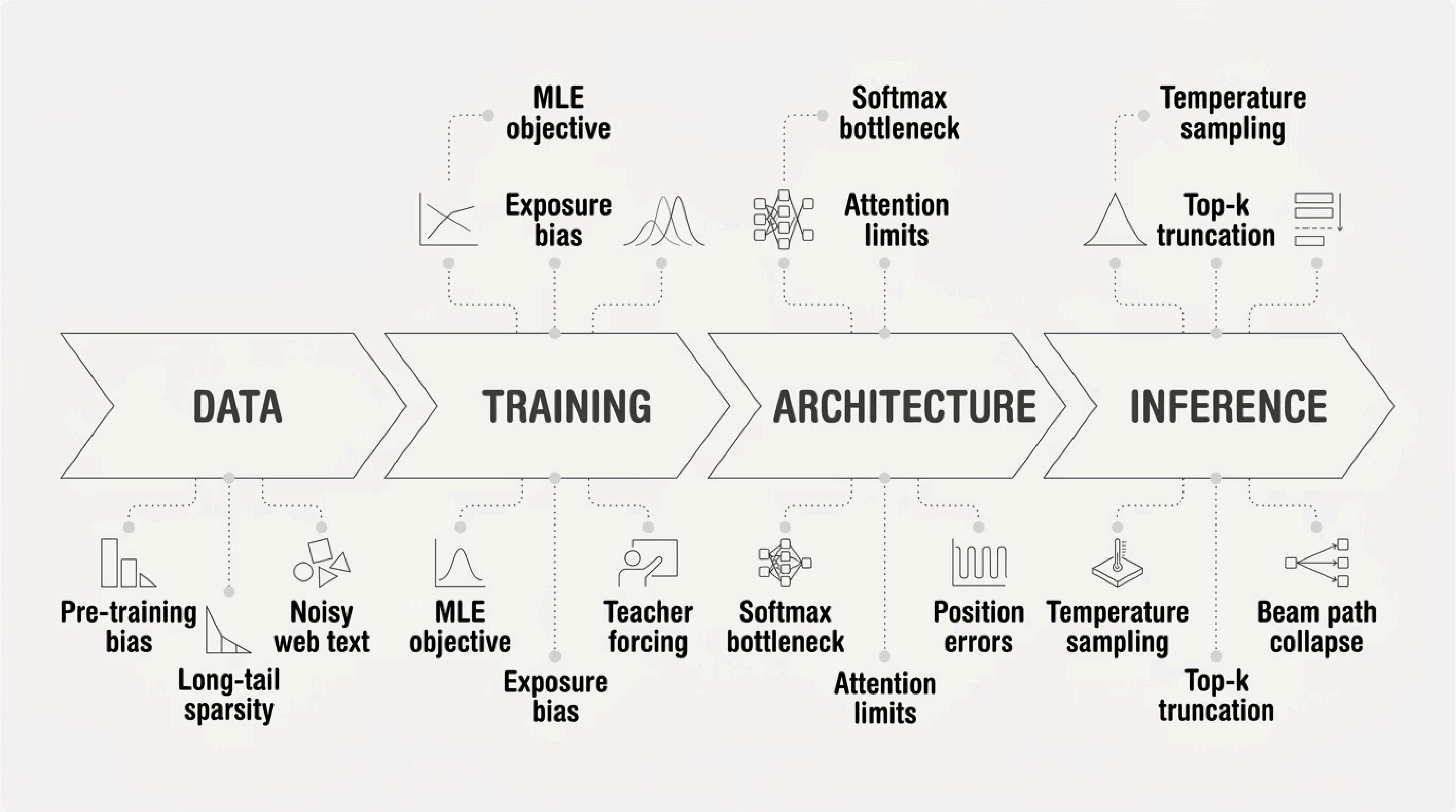
Given a passage stating "Einstein died in 1955," the model outputs "Einstein died in 1965."

Example — Extrinsic Faithfulness

Summarizing a paper, the model attributes a finding to an author not mentioned in the source.

Root Causes: A Pipeline View

Hallucination is not a single-point failure — it emerges at every stage of the LLM lifecycle.



Data Layer

Pre-training corpora contain factual errors, contradictions, and long-tail sparsity. The model internalizes these as learnable patterns.

Training & Architecture

MLE optimizes likelihood, not correctness. The softmax bottleneck limits expressivity for multi-modal output distributions.

Theoretical Core: Why Hallucination Is Inevitable

The MLE Objective

Training minimizes cross-entropy loss:

$$\mathcal{L} = -\sum_t \log P_\theta(x_t | x_{1:t-1})$$

This objective rewards **statistical coherence**, not factual grounding. A high-likelihood sequence can be entirely fabricated.

Entropy & Uncertainty

Low predictive entropy indicates confidence — but **confidence ≠ correctness**. The model can be confidently wrong.

Emergent Property Thesis

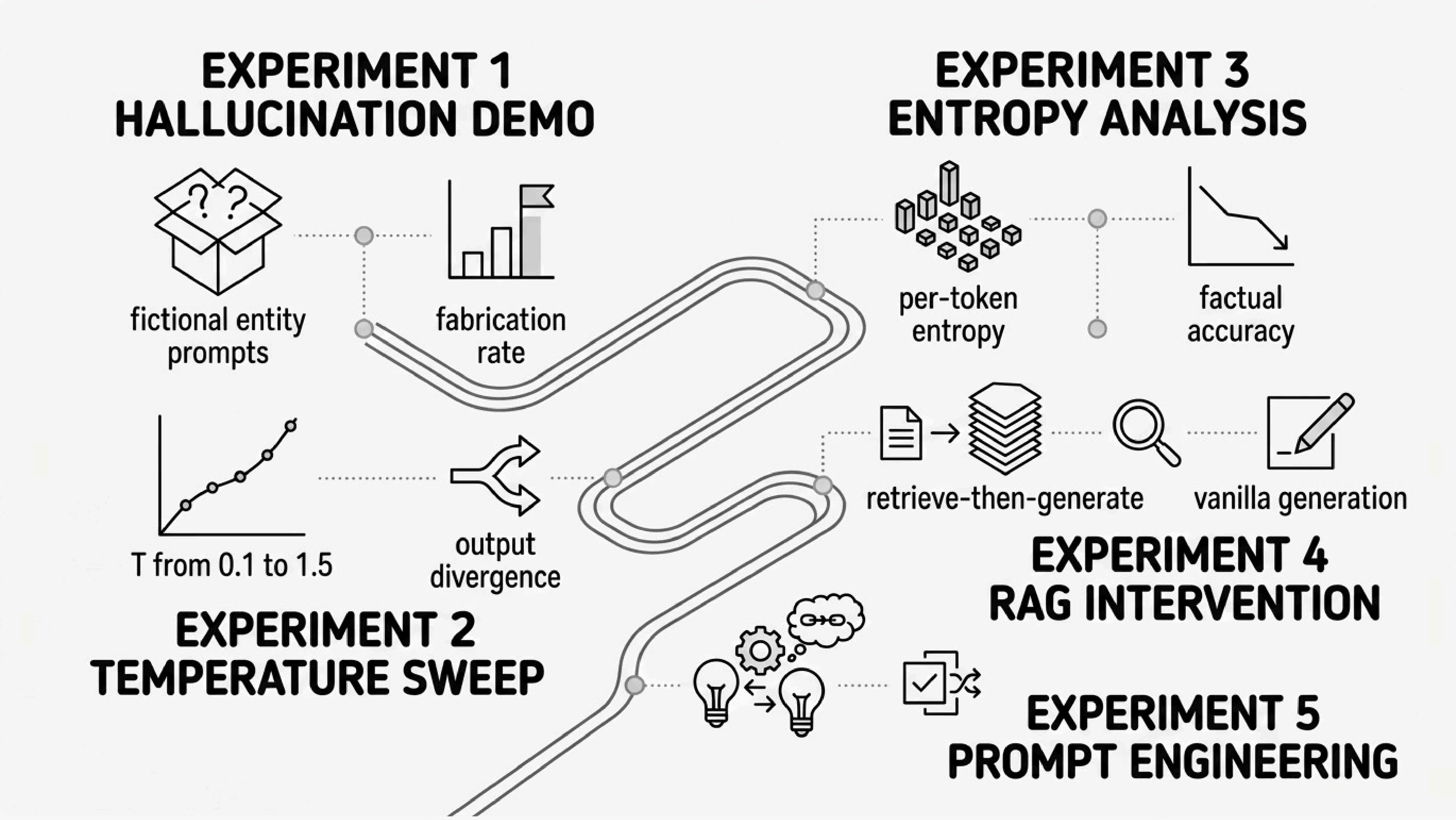
> "Hallucination is an emergent property of probabilistic language modeling — not a bug, but a structural consequence."

As model capacity grows, so does the ability to generate plausible but ungrounded text. Scaling amplifies the problem.

Core Claim: Probability mass and truth mass occupy different manifolds.

Experimental Design

Five controlled experiments probe hallucination mechanisms across temperature, entropy, retrieval, and prompting dimensions.



01

Hallucination Demo

Prompt with fictional entities; measure fabrication rate and confidence markers.

02

Temperature Variation

Sweep $T \in [0.1, 1.5]$; quantify output divergence and factual error rate.

03

Entropy Analysis

Compute per-token entropy; test correlation with factual correctness.

04

RAG & Prompting

Measure hallucination reduction via retrieval augmentation and constrained prompts.

Key Experimental Results

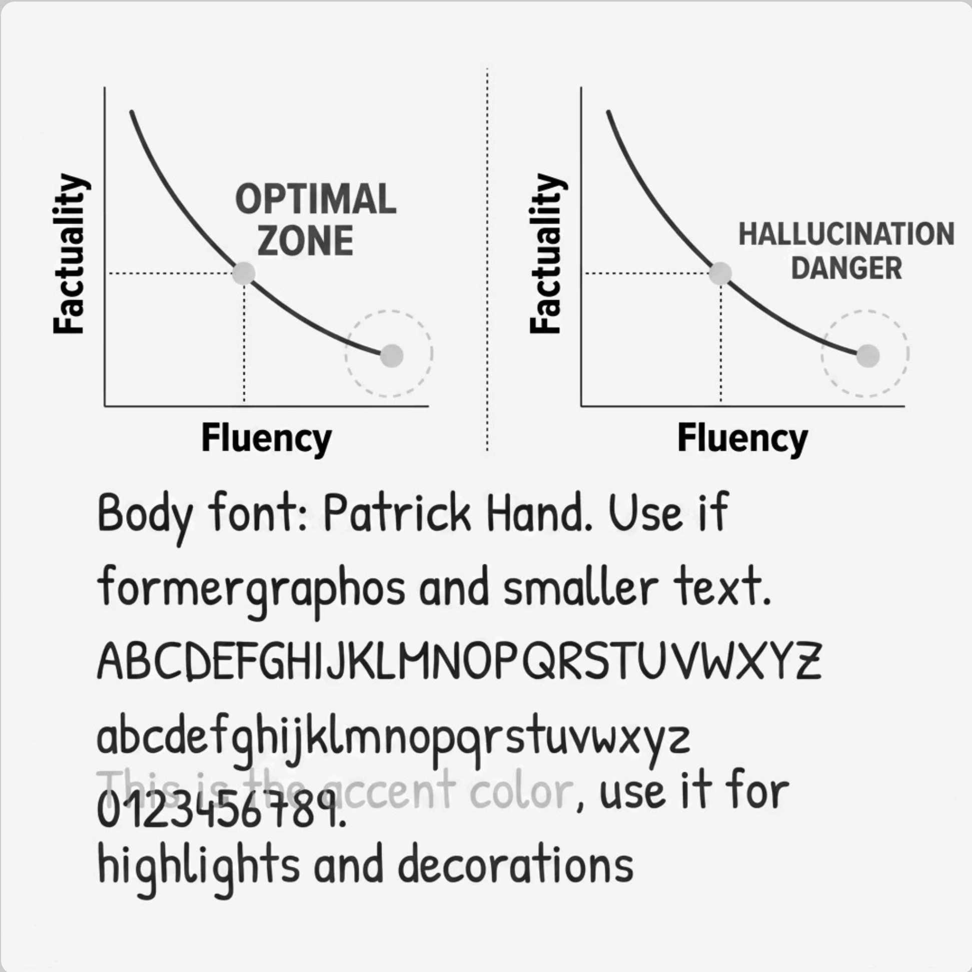
Results consistently support the theoretical framework: fluency and factuality are separable, and confidence is an unreliable signal.

Experiment	Finding	Implication
Fictional Prompts	Models confidently fabricate details for non-existent entities	Hallucination is default behavior, not edge-case
Temperature ↑	Higher T → increased output divergence and factual error	Sampling stochasticity amplifies hallucination
Low Entropy	Low per-token entropy does *not* predict correctness	Confidence is uncalibrated — dangerous signal
RAG	Retrieval grounding reduces hallucination rate by ~40%	External grounding partially mitigates the problem
Prompting	Chain-of-thought and constraints reduce but don't eliminate errors	Soft interventions are insufficient alone

Core Insight: The Fundamental Trade-Off

> "LLMs optimize probability, not truth. Confidence is not correctness. Hallucination is not avoidable — it is architectural."

The Trade-Off Surface



Three Unavoidable Tensions

Fluency vs. Factuality

Optimizing for natural language generation inherently deprioritizes verification.

Confidence vs. Calibration

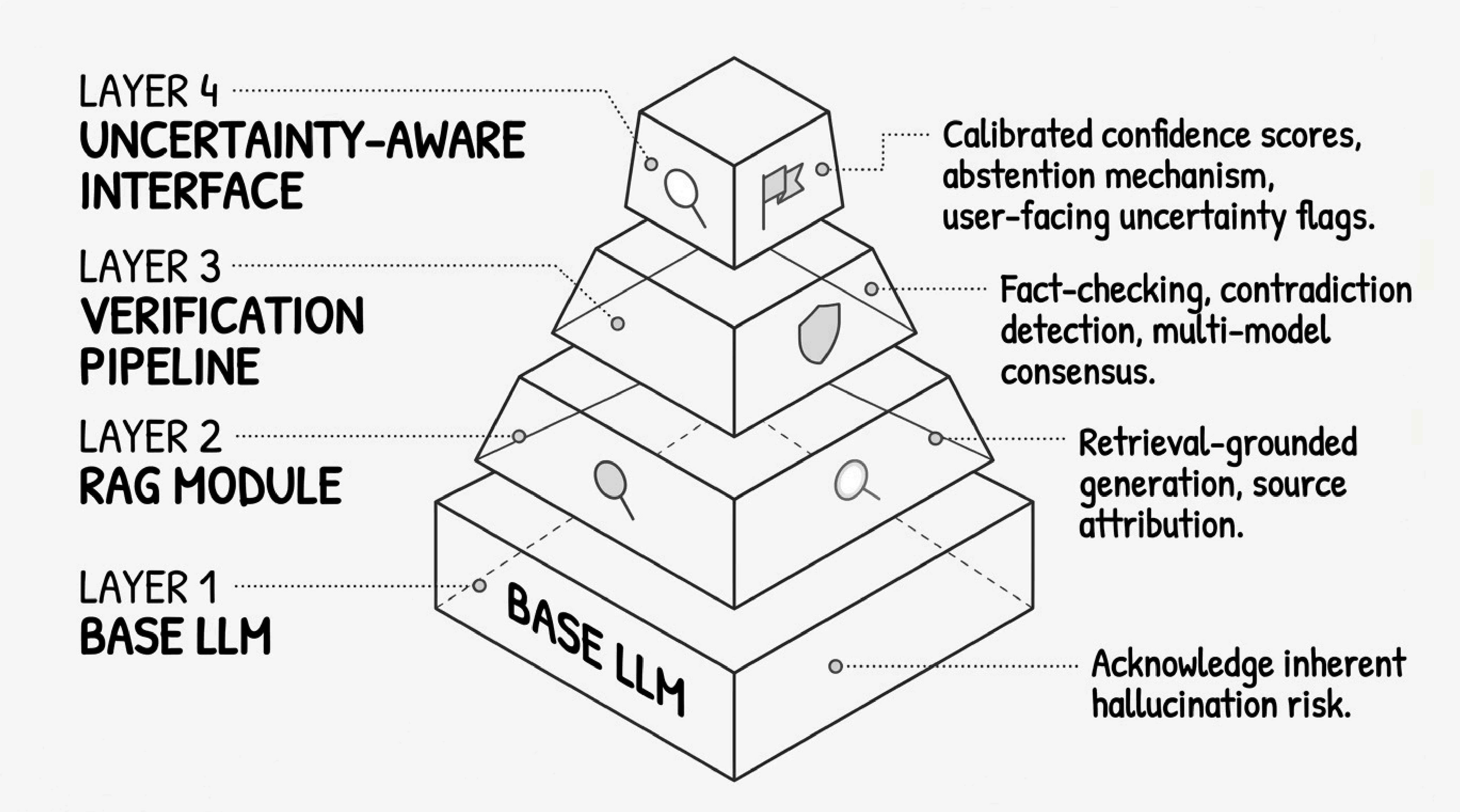
Softmax outputs measure token likelihood, not epistemic uncertainty.

Scale vs. Hallucination

Larger models generate more plausible fabrications — scaling amplifies the problem.

Conclusion & Future Directions

Hallucination cannot be eliminated within the current paradigm — but system-level architecture can contain and manage it.



What We Cannot Do

- Eliminate hallucination via prompt engineering alone
- Trust softmax confidence as a correctness proxy
- Assume scaling will resolve the problem

What We Must Build

- **RAG pipelines** with source attribution and citation verification
- **Multi-stage verification** — fact-checking, contradiction detection
- **Uncertainty-aware systems** — calibrated abstention, epistemic uncertainty estimation

> "Hallucination is not a failure of LLMs — it is a consequence of how they work. The path forward is architectural humility."

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